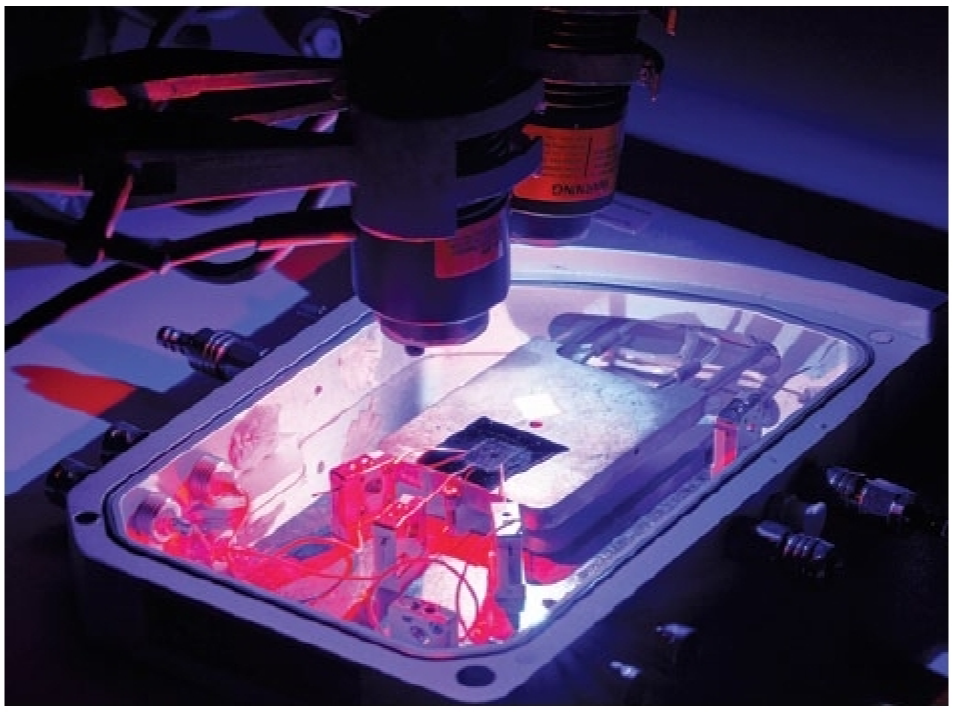
Tech News

Technology Updates

Electronic chip mimics the brain to make memories in a flash



Researchers from Functional Materials and Microsystems Research Group at RMIT have demonstrated that the chip can perform logic operations—information processing—ticking another box for brain-like functionality (Credit: RMIT University)

Researchers from RMIT University, after drawing inspiration from optogenetics, have developed a device that replicates the way the brain stores and loses information.

The new chip is based on an ultra-thin material that changes electrical resistance in response to different wavelengths of light, enabling it to mimic the way that neurons work to store and delete information in the brain.

Research team leader Dr Sumeet Walia says that the technology moves us closer towards artificial intelligence (AI) that can harness the brain's full sophisticated functionality. "Our optogenetically-inspired chip imitates the fundamental biology of nature's best computer—the human brain," he explains.

Dr Walia adds, "Being able to store, delete and process information is critical for computing, and the brain does this extremely efficiently. We were able to simulate the brain's neural approach simply by shining different colours onto the chip.

"This technology takes us further on the path towards fast, efficient and secure light-based computing. It also brings us a step closer to the realisation of a bionic brain—a brain-on-a-chip that can learn from its environment just like humans do."

According to Dr Taimur Ahmed, lead author of the study, being able to replicate neural behaviour on an artificial chip offers exciting avenues for research across sectors. He says, "This technology creates tremendous opportunities for researchers to better understand the brain and how it is affected by disorders that disrupt neural connections, like Alzheimer's disease and dementia."

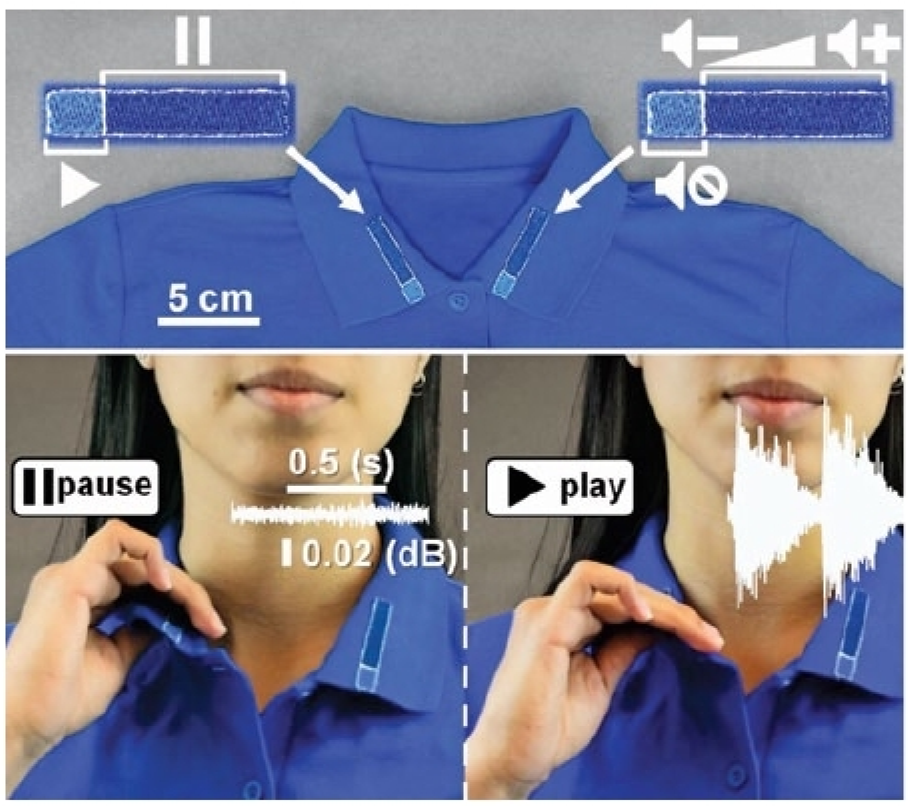
Clothing that lets users turn on electronics while turning away bacteria

Researchers at Purdue University have developed a new fabric that allows wearers to control electronic devices through clothing.

According to Ramses Martinez, assistant professor in School of Industrial Engineering and in Weldon School of Biomedical Engineering in Purdue's College of Engineering, this is the first technique that can transform an existing cloth item or textile into a self-powered e-textile containing sensors, music players or simple illumination displays using simple embroidery without the need for expensive fabrication processes that require complex steps or expensive equipment.

He says, "These self-powered e-textiles also constitute an important advancement in the development of wearable machine-human interfaces, which now can be washed many times in a conventional washing machine without apparent degradation."

Purdue waterproof, breathable and antibacterial self-powered clothing is based on omniphobic triboelectric nanogenerators (RF-TENGs), which use simple embroidery and fluorinated molecules to embed small electronic components and turn a piece of clothing into a mechanism for powering devices. The team says the RF-TENG technology is like having a wearable remote control that also keeps odours, rain, stains and bacteria away.



Purdue waterproof, breathable and antibacterial self-powered clothing is based on omniphobic triboelectric nanogenerators (Credit: Purdue University)